

Notation.

Let $X: U \rightarrow \mathbb{R}^3$ be a patch. Define

$$g_{i\bar{j}} \stackrel{\text{def}}{=} \langle X_i, X_{\bar{j}} \rangle$$

So that given two tangent vectors $A = \sum a^i X_i$ and $B = \sum b^{\bar{j}} X_{\bar{j}}$ exists

$$\langle A, B \rangle = \sum_{1 \leq i, \bar{j} \leq 2} a^i b^{\bar{j}} \cdot \langle X_i, X_{\bar{j}} \rangle = \begin{pmatrix} b^1 & b^2 \end{pmatrix} \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{pmatrix} a^1 \\ a^2 \end{pmatrix}$$

And, $g \stackrel{\text{def}}{=} \det(g_{i\bar{j}})$ and $g^{kl} = (k, l)$ entry of $(g_{i\bar{j}})^{-1}$.

The normal of X at $X(s)$ is ($s = (a, b)$)

$$n(s) = \frac{X_1(s) \times X_2(s)}{\|X_1(s) \times X_2(s)\|}.$$

And, the second-derivatives at $X(s)$ is

$$X_{i\bar{j}}(s) = \frac{\partial^2 X}{\partial u^i \partial u^{\bar{j}}}(s)$$

And, the intrinsic normal of a unit speed curve in the image of X is

$$S = n \times T.$$

The Normal and Geodesic Curvature of a unit speed curve γ in X

$$K_n = \langle \gamma''(s), n(\gamma(s), \gamma^2(s)) \rangle \text{ and } K_g = \langle \gamma''(s), S(s) \rangle$$

The coefficients of the second fundamental form of X is

$$L_{i\bar{j}} = \langle X_{i\bar{j}}, n \rangle \quad (1 \leq i, \bar{j} \leq 2)$$

The Christoffel symbols

$$\Gamma_{i\bar{j}}^k \stackrel{\text{def}}{=} \sum_{l=1}^2 \langle X_{i\bar{j}}, X_l \rangle g^{lk}.$$

The Second-fundamental form II on M is the bilinear form

$$II: T_p M \times T_p M \rightarrow \mathbb{R}; (X, Y) \mapsto \sum_{1 \leq i, \bar{j} \leq 2} L_{i\bar{j}} X^i \cdot Y^{\bar{j}}$$

with associated Matrix L . ($L_k^l = \sum L_{i\bar{k}} g^{i\bar{l}}$)

